

On Scope Classification and Current Knowledge-Editing Benchmarks: A Negative Result, with INLAY as a Gradient-Free Case Study

Anonymous ACL submission

Abstract

Every memory-based knowledge editor in the SERAC lineage depends on a **scope decision**: given a query, does a stored edit apply? We report that current knowledge-editing benchmarks cannot measure this decision at all. Using INLAY, a gradient-free editor we built to obtain exact per-query ground truth — the model is frozen, edits live in an external addressable memory, and applying an edit is a bias added along one token’s unembedding direction at decode time — we execute *every* candidate router action on 1,689 queries spanning three datasets and three input conditions. An oracle router choosing the best action every time ties a one-line static policy to four decimal places in *all nine* dataset-by-condition cells: the maximum attainable gain of any per-query routing method is 0.00 points. Abstinence is the sole winning action zero times out of 1,689. The cause is structural: these are counterfactual benchmarks whose evaluation question asks for the post-edit answer, so answering from parametric knowledge is wrong by construction, and a benchmark without negatives cannot reward a classifier’s ability to reject. This generalizes beyond our system to the whole scope-classifier family the benchmarks are used to evaluate. We also report where INLAY itself does not win — WISE beats it on Qwen2.5-7B CounterFact, and retrieval-augmented generation beats every method we tested, INLAY included, on rigorously matched RippleEdits — and disclose two bugs found during a self-audit of our own routing machinery, neither of which changed a published headline number outside noise.

1 Introduction

A language model’s knowledge is not stored anywhere one can point to. There is no row reading “the CEO of X is Y ”; the fact exists as a pattern distributed over billions of weights, each of which also participates in thousands of unrelated

facts. Correcting one fact after training — because it changed in the world, or was wrong to begin with — is the knowledge editing problem, and a large family of methods now addresses it by locating weights and rewriting them (Meng et al., 2022, 2023; Fang et al., 2025) or by storing corrections outside the weights and deciding, per query, whether a stored correction applies (Mitchell et al., 2022b; Hartvigsen et al., 2023; Wang et al., 2024).

That second family is the subject of this paper. Every method in it needs a **scope classifier**: a component that looks at an incoming query and decides whether an edit governs it, an in-context demonstration should fire, or the model should simply answer from what it already knows. SERAC’s scope classifier (Mitchell et al., 2022b) is the namesake instance, but the same decision recurs under different names in every architecture that pairs a frozen or lightly-modified base model with an external correction mechanism. We built one such system, executed every action it could take on every query in three standard benchmarks, and found that the decision this entire architecture family is built around cannot be rewarded by the benchmarks used to evaluate it.

Why we can say this with certainty rather than suspicion. The usual evidence for “the benchmark doesn’t test X ” is indirect: an ablation that shouldn’t matter doesn’t, or a simpler baseline is competitive. We instead computed the *ceiling*. For 1,689 queries across CounterFact, WikiUpdate, and MQuAKE-CF in structured, unstructured, and extracted-evidence conditions, we executed *every* candidate action (recite the stored edit, reason over retrieved evidence, or abstain) and scored every outcome, giving per-query ground truth for which action is correct. An oracle router that always picks the best available action is *exactly* equal to a one-line static policy in all nine cells (§6). There is no headroom for any router to

find, because abstention — the action a scope classifier exists to choose — is the sole winning action zero times.

Where INLAY fits. To get that ground truth we needed a system whose every action could actually be executed and scored under identical conditions, including a gradient-free recitation path with no per-edit optimization cost. We built INLAY: the base model stays frozen, each edit is a row in an external addressable memory (a semantic key plus the answer’s token sequence), and applying an edit adds a logit-space bias along the answer’s unembedding direction at decode time. Writing an edit is a table insertion, about 5–15 ms with no gradient step; deleting an edit is removing the row; and when no edit fires, output is unchanged bit for bit. INLAY is competitive with or ahead of gradient-based editors on direct single-fact editing (§5), but it is not always the strongest method we tested, and we report the cases where it loses plainly (§5, §5.4). INLAY is the vehicle for this paper’s central claim, not the claim itself.

Contributions. (i) A negative result with a measured ceiling, not a suspicion: on 1,689 queries with per-action ground truth, an oracle router ties a one-line static policy exactly in every cell, and abstention is never uniquely correct (§6–§7). (ii) INLAY, a gradient-free editor with exact deletion and locality by construction, evaluated across four model families and all three AKEW input conditions, including evidence formats for which weight editing has no mechanism (§3–§5). (iii) An honest accounting of where INLAY loses: WISE beats it on Qwen2.5-7B CounterFact, and RAG beats every method on rigorously matched RippleEdits (§5, §5.4) — correcting an assumption implicit in earlier drafts of this work that the named boundary was against weight editors, when the benchmark that actually exposes it favors retrieval instead. (iv) A transparent self-audit of our own routing machinery, including a gate-bypass bug that made a measured mitigation inactive in the runs behind our own published numbers (§8). (v) A diagnosis and fix for multi-hop chain termination, more than tripling accuracy and holding at two model scales (§5.3).

2 Related Work

Weight editing. ROME (Meng et al., 2022) treats a mid-layer MLP as a linear associa-

tive memory and solves for a rank-one update; MEMIT (Meng et al., 2023) spreads many updates across several layers for mass editing; AlphaEdit (Fang et al., 2025) projects the update into the null space of preserved knowledge. All three require a clean (prompt, target) pair and mutate shared parameters. Hase et al. (2023) show that editing at layers causal tracing does *not* highlight often works equally well, which weakens the inference from localization to edit site.

Memory-based and scope-classified editing. SERAC (Mitchell et al., 2022b) introduces a scope classifier deciding whether an edit applies, routing in-scope inputs to a counterfactual model. MEND (Mitchell et al., 2022a) learns a gradient transform. IKE (Zheng et al., 2023) shows few-shot demonstrations can teach a model to prefer an injected fact over its parametric belief. GRACE (Hartvigsen et al., 2023) and WISE (Wang et al., 2024) add discrete codebooks and side memories respectively, each with their own routing decision at generation time. **We build directly on SERAC’s scope-classifier idea and IKE’s demonstration mechanism; neither is claimed as novel here.** Our addressing follows product-key memory (Lample et al., 2019) and nearest-neighbour language models (Khandelwal et al., 2020).

Benchmarks. CounterFact and zsRE evaluate single-fact edits; RippleEdits (Cohen et al., 2024) adds entailed consequences; MQuAKE (Zhong et al., 2023) adds multi-hop chains; AKEW (Wu et al., 2024) supplies unstructured and extracted evidence alongside clean triples. §6–§7 is a claim about what this entire family of benchmarks can and cannot measure, independent of which scope-classified method is being scored on them.

3 Method: INLAY

INLAY exists to make the routing question in §6 answerable with exact, executable ground truth: every candidate action must be a real code path we can run and score, including a gradient-free recitation path cheap enough to execute at scale. We describe it briefly here; it is the vehicle for this paper’s argument, not its subject.

3.1 Design

The model is frozen. Each edit is a row holding a key (an embedding of the fact’s question form),

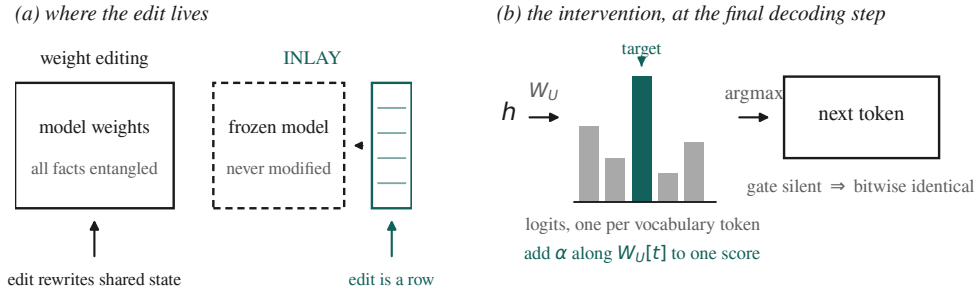


Figure 1: **(a)** Weight editing rewrites state shared by every fact. INLAY leaves the model frozen and appends a row to a table beside it. **(b)** The intervention is a bias along one token’s unembedding direction at the final decoding step, so it moves exactly one score. When the gate does not fire, nothing is added and output is bitwise identical.

the answer’s token sequence, and metadata (Figure 1a). **Writing is insertion**, not optimization — no gradients, no covariance statistics, measured at 5–15 ms. **Deletion is exact**: removing the row removes the edit, which matters when removal is a regulatory obligation rather than a nicety (see the Ethics Statement). **Edits cannot interfere**: they are physically separate state.

3.2 Addressing

Keys come from a sentence encoder (Reimers and Gurevych, 2019) rather than the base model’s hidden states: it is fast, its geometry is purpose-trained for paraphrase matching, and it stays fixed if the base model is swapped. Keys are compressed with a Johnson–Lindenstrauss random projection (Johnson and Lindenstrauss, 1984), chosen over a learned projection because a learned one would drift as the edit distribution changed and would silently invalidate every stored key; a fixed random matrix never invalidates the memory, at the cost of a few points of distortion.

3.3 Gating and playback

Retrieval always returns *something*. A gate stacks an absolute similarity threshold, a margin over the runner-up, and a relation-residual check asking whether the query concerns the stored relation rather than merely the stored subject (we return to the reliability of this gate in §8). When the gate fires, playback walks the stored token sequence and, at each decoding step, adds a bias along that token’s unembedding column so it wins the argmax (Figure 1b). Positions outside the answer span are untouched. Intervening in logit space rather than hidden state is a safety argument: a logit bias moves exactly one score, while a hidden-state edit propagates through the unembedding to every token’s logit in ways that are hard to bound;

and because the intervention is gated, a silent gate leaves behaviour bitwise unchanged.

4 Experimental Setup

Base models: GPT-2-XL (1.5B), GPT-J-6B, Qwen2.5-7B, and Mistral-7B-v0.3. Benchmarks: CounterFact, zsRE, RippleEdits (Cohen et al., 2024), and AKEW (Wu et al., 2024) over CounterFact, WikiUpdate and MQuAKE-CF in structured, unstructured and extracted conditions. Baselines run through EasyEdit (Wang et al., 2023).

Two commitments are worth stating because they cost us results. Splits are **subject-disjoint**: a subject never crosses a train/test boundary, so a learned component cannot memorize an entity and appear to generalize. Scoring uses **one convention everywhere** — diacritic- and case-insensitive substring match against gold and its aliases — rather than each method’s internal metric, so numbers are comparable across methods that report differently.

One implementation detail materially affects correctness: calling `BaseEditor.edit` with `sequential_edit=False` restores weights before returning, so generating after it returns scores the *unedited* model. All runs reported here use `sequential_edit=True` with an explicit state-dict snapshot and restore under harness control. This bug, and a second one that specifically invalidated an earlier round of matched RippleEdits numbers, are discussed in §8.

5 Results

5.1 Edit accuracy and write cost

Figure 2 gives the GPT-J-6B comparison. AlphaEdit is the strongest weight editor at 89.12%, consistent with its null-space constraint making each update more surgical. ROME and MEMIT tie at 83.67%: single-fact CounterFact is where

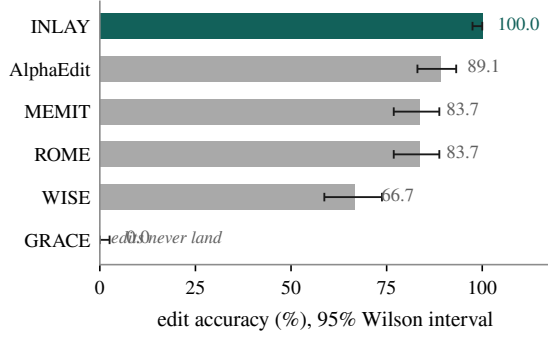


Figure 2: Edit accuracy on CounterFact, structured input, GPT-J-6B, $n = 147$, with 95% Wilson intervals (Wilson, 1927). Scored with a single convention rather than each method’s internal metric.

both converge, and MEMIT’s advantage is multi-edit preservation, not single-edit efficacy. Two results are failures of methods that work elsewhere: WISE reaches 66.67% despite EasyEdit’s own post-edit metric reporting the edit landed on 141 of 147 examples, because WISE routes through a side-memory module at generation time and that routing is an additional failure point in-place updates lack; GRACE reads exactly 0.0 on every edit, its radius-based key rarely firing on paraphrases. On the harmonic-mean score across ES/PS/NS at $N=2000$, INLAY leads at 0.8926 against ROME’s 0.797, WISE’s 0.703, AlphaEdit’s 0.4595, and MEMIT’s 0.4314 (full-results-audit.md, Part A2). INLAY writes in 5–15ms with no gradient step against seconds for gradient-based editors, a ratio of roughly 1600×

This ranking does not hold on every model.

On Qwen2.5-7B CounterFact at $N=2000$, WISE reaches a harmonic-mean score of **0.9466** (ES 1.0, PS 0.8553, NS 1.0) against INLAY’s **0.8944** at $N=5000$ (ES 1.0, PS 0.8776, NS 0.8232) — *WISE wins on Qwen*, the one place in our CounterFact results where INLAY is not the strongest method. We report this plainly rather than selecting the model on which INLAY leads. MEMIT and AlphaEdit’s Qwen numbers in the source logs use a different, non-comparable “dual-metric” probability-of-success framing rather than the harmonic-mean convention used for INLAY/ROME/WISE/GRACE, so they are not placed in this ranking; see Appendix A for both metric families side by side, flagged as such.

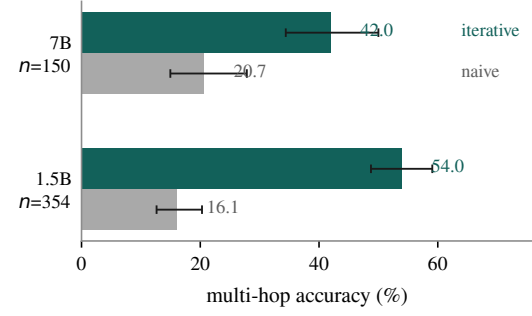


Figure 3: Iterative decomposition versus naive single-shot on MQuAKE-CF after the per-hop fallback fix, with 95% Wilson intervals.

5.2 Non-structured evidence

Weight-editing methods need a (prompt, target) pair to compute an update from. They have no mechanism for “here is a paragraph of evidence, install whatever matters,” so on AKEW’s unstructured and extracted conditions they are not merely worse but inapplicable; we state this as a structural limitation rather than working around it. INLAY handles all three conditions. Accuracy is lower on extracted input (78.23%) than on unstructured prose (87.07%) despite retrieval finding the correct card 98.64% of the time in both, which locates the loss in extraction noise rather than retrieval.

5.3 Multi-hop

Our first iterative loop scored 5.0% against a naive single-shot baseline’s 22.5% — four times *worse* than doing nothing, with every sampled failure producing no answer. The cause was an assumption: MQuAKE-CF chains mix edited facts with ordinary unedited world facts (277 of 354 groups contain exactly one edit across a two- or three-hop chain), and the loop treated every hop as requiring retrieval against the edit index, so a hop needing a never-edited fact terminated the chain answerless. Treating a retrieval miss as a signal to answer that hop from parametric knowledge and *continue* raised accuracy to 47.5% on the same sample, and it holds at scale (Figure 3).

5.4 Compositional propagation: RippleEdits

RippleEdits tests whether an edit’s entailed consequences propagate while unrelated same-subject facts are preserved — a compositional axis that single-fact CounterFact and zsRE cannot probe. An early, less rigorous comparison (base/RAG/INLAY sharing an identical query set)

Method	GPT-J-6B	Qwen2.5-7B
in-context (RAG)	0.3964	0.4381
WISE [†]	0.3425	0.1421
INLAY	0.2253	0.2871
AlphaEdit	0.1409	0.1683
ROME	0.1314	0.1453
base	0.0740	0.1522

Table 1: Matched-manifest RippleEdits (identical wikidata-verified subjects across every method, $n=100$ edits, generation-based scoring, edit-applied verified for every run). [†] WISE runs in its native sequential-accumulating mode; every other method is single-edit isolated with a per-edit weight restore. Source: the audit addendum (2026-08-19), §B.

put INLAY’s preservation at 0.049, the worst of the methods compared, and this number was the basis for an earlier framing of INLAY’s boundary as a loss to weight-editing methods. **That framing does not survive a rigorous comparison, and we correct it here.**

Table 1 gives the corrected, matched-manifest protocol: identical wikidata-verified subjects for every method on both models, generation-based scoring, and edit application independently verified rather than assumed. **Retrieval-augmented generation leads every method we tested on both models** — 0.3964 on GPT-J-6B and 0.4381 on Qwen2.5-7B, ahead of every gradient-free and weight-editing method alike. INLAY does not beat this baseline (0.2253 / 0.2871). **The paper’s honest limitation is therefore that INLAY loses to RAG, not that it loses to the weight-editing family:** on Qwen2.5-7B, INLAY’s 0.2871 beats every weight/adaptor editor we ran (ROME 0.1453, AlphaEdit 0.1683, WISE 0.1421), and on GPT-J-6B it beats ROME and AlphaEdit while trailing WISE’s side-memory accumulation mode. The mechanism behind INLAY’s own weakness here is unchanged from what we reported previously: scope precision on hard negatives (same-subject, different-relation queries) is the open problem, and it is exactly where a benchmark with negatives, unlike CounterFact or zsRE, can see it.

5.5 Sequential editing: naming which method collapses

An earlier draft of this work described weight editors as compounding damage under repeated editing, as a general property of the family. That claim is too broad. On GPT-2-XL, ROME’s retention and locality both

collapse to 0.0 by $n=25$ sequential edits (Part D of full-results-audit.md), and the same collapse reproduces on GPT-J-6B (seq_rome_gptj.log). **MEMIT does not exhibit this failure mode on GPT-J-6B:** it holds retention and locality at 1.0 through the recorded checkpoints in the same sequential test (seq_memit_gptj.log), consistent with MEMIT’s design goal of spreading updates across layers specifically to support mass editing. **We narrow the claim to name ROME specifically:** weight editing is not inherently unstable under repeated edits, and a method built for mass editing (MEMIT) does not show the same failure a method built for surgical single-edit updates (ROME) does.

6 The Limits of Routing on Current Benchmarks

Every method in the scope-classified family includes a **scope decision**: given a query, does a stored edit apply, and what should be done about it? We built increasingly careful machinery for this decision, measured statistically significant gains from it, and then found the gains were not what they appeared to be.

6.1 A second-order signal

Gating was net-negative on MQuAKE-CF in every input condition, by up to 15.87 points. Two natural fixes failed diagnostically: recalibrating the verifier threshold moved the false-fire rate from 18.33% to 18.22%, and raising the confidence threshold from 0.85 to 0.97 changed *nothing* — byte-identical decisions, because confidence on the *wrong* retrievals already exceeded 0.97. The router was asking how confident the verifier was, when it needed to know whether that confidence was *discriminative*. A verifier scoring 0.99 on the top candidate and 0.98 on four unrelated ones is saturated, not confident, and no threshold on a top-1 score can see that.

We trained a head predicting retrieval reliability from the *shape* of the candidate set — margins to the next best candidate, how many candidates clear threshold, neighborhood entropy, subject diversity — on CounterFact and WikiUpdate with MQuAKE-CF held out entirely. It reached 0.956 AUROC on the held-out dataset; 71.5% of coefficient mass sat on margin features, and the largest single weight was -1.43 on the best *com-*

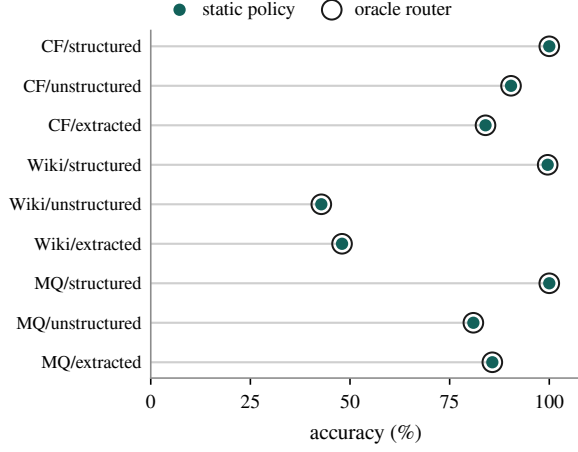


Figure 4: Every candidate action executed and scored on 1,689 queries (CF = CounterFact, MQ = MQuAKE-CF). The oracle ring is concentric with the static-policy point in all nine cells: the best attainable router and one line of code are indistinguishable.

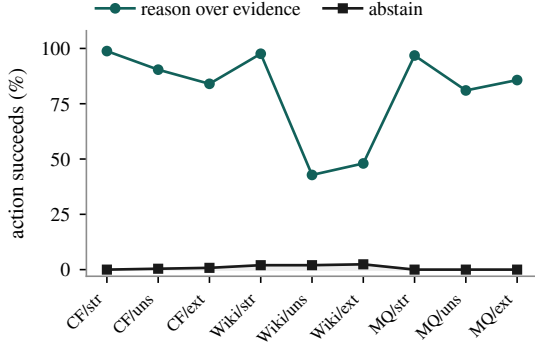


Figure 5: Per-action success rates. Abstention lies against the floor in every cell, which is the mechanism behind the zero headroom in Figure 4.

peting candidate’s score. Used adaptively it produced +15.9 points on two MQuAKE-CF cells ($p=0.0019$, Holm-corrected (Holm, 1979)), with zero discordant pairs where fixed gating already worked.

6.2 Measuring the ceiling

Rather than continue improving the router, we measured its ceiling. For 1,689 queries across all three datasets and conditions we executed *every* candidate action and scored the result, yielding per-query ground truth for which actions work.

An oracle router choosing the best available action on every query is exactly equal to a one-line static policy — recite directly where legal, otherwise reason over the retrieved evidence — to four decimal places in all nine cells (Figure 4, Table 2). The maximum attainable gain of any per-query routing method over one line of code is 0.00

Dataset	Condition	n	Static	Oracle
CounterFact	structured	250	1.0000	1.0000
CounterFact	unstructured	250	0.9040	0.9040
CounterFact	extracted	250	0.8400	0.8400
WikiUpdate	structured	250	0.9960	0.9960
WikiUpdate	unstructured	250	0.4280	0.4280
WikiUpdate	extracted	250	0.4800	0.4800
MQuAKE-CF	structured	63	1.0000	1.0000
MQuAKE-CF	unstructured	63	0.8095	0.8095
MQuAKE-CF	extracted	63	0.8571	0.8571
Pooled		1689	0.7874	0.7874

Table 2: Headroom is +0.0000 in every cell. “Static” is one line of code with no model and no training; “Oracle” is an upper bound no real router can exceed.

points.

The mechanism is stark (Figure 5). Abstention succeeded on 19 of 1,689 queries, and on all 19 of those, reasoning or direct recitation succeeded too. **Abstention is the sole winning action zero times.**

6.3 What this does to the preceding subsection

The gains in §6.1 are real and we stand behind the numbers. The *explanation* was wrong. We framed the head as adapting per query to the local regime; it does not, because there is no per-query decision worth making. What it does is suppress gates that misfire — it converges toward the static policy. Its gains measure the harm the fixed gates were doing, not insight it adds. A one-line rule matches it everywhere, with no training data and none of the k -fold cross-encoder cost.

7 Why Abstention Cannot Be Measured Here

The reason is structural. These are **counterfactual** benchmarks: the evaluation question asks for the *post-edit* answer, so answering from parametric knowledge returns the *pre-edit* value, which is wrong by construction. No query in the suite has “no edit applies, answer from what you know” as its correct behaviour.

This generalizes beyond our system. Scope classification is load-bearing for the SERAC-descendant family (Mitchell et al., 2022b), whose premise is a learned decision about whether a stored edit applies. On these benchmarks that decision is degenerate: the answer is always “yes.” **A benchmark without negatives cannot measure a classifier’s ability to reject**, and an

abstention path evaluated on one can only lose points.

It also explains the RippleEdits result of §5.4. RippleEdits *does* contain same-subject, different-relation queries and therefore partial negatives, and it is exactly there that INLAY’s preservation weakness shows up and every method’s aggregate score drops relative to CounterFact and zsRE. Scope precision matters exactly where a benchmark has negatives, which is precisely where the main single-fact benchmarks are silent.

The missing condition is constructible: ask evaluation questions against an index that does *not* contain the corresponding edit. On such queries abstention becomes correct, the oracle should separate from the static policy, and routing acquires measurable headroom for the first time. We did not build this benchmark for the present paper; we flag it as the concrete, falsifiable next step this result implies.

8 Auditing Our Own Machinery

A negative result about benchmark design is only as credible as the honesty of the system used to produce it. We audited INLAY’s own gating machinery after submission-quality numbers were already in hand and report two findings that could have changed those numbers, together with what we did to check whether they did.

The margin gate is a no-op in the protocol behind our headline numbers. The margin-over-runner-up gate is a genuine, measured locality fix at scale in a multi-slot regime: on a 400-edit sequential test on GPT-2-XL, sweeping the margin from 0 to 0.15 moves locality from 0.5667 to 1.0 (score 0.7221 $\rightarrow$ 0.9975). But every headline CounterFact and zsRE number in §5.1 is produced under a single-edit-per-example protocol, where memory is cleared before every write and only one slot is ever occupied. In that regime the margin gate has nothing to compete against and is a no-op; it only matters in multi-slot regimes like the sequential test that measured it. We state this so the margin gate’s benefit is not read into results it did not affect.

A gate-bypass bug meant the relation gate was never active in the runs behind our published matched-RippleEdits numbers. A code audit found that `answer_playback()` in `gpt2_memory_semkey.py` — the function

Aggregate	Published (bug)	Corrected	Δ
GPT-J-6B	0.2253	0.2283	+0.0030
Qwen2.5-7B	0.2871	0.2891	+0.0020

Table 3: Validation rerun at `rel_gate=0.2`, INLAY only, matched manifest, with the gate-bypass bug fixed. Source: the audit addendum (2026-08-19), §D2.

every real generation path runs through, including RippleEdits — checked only the absolute similarity score and silently skipped the margin and relation-residual gates that the scoring-only code path already supported. The relation gate’s measured over-firing reduction (over-fire rate 0.9667 $\rightarrow$ 0.6667 at `rel_gate=0.2`) was therefore validated only on a separate teacher-forced harness and was **never actually active** in the matched-RippleEdits run that produced the published preservation numbers in Table 1. We fixed this by adding a single shared routing decision used identically by both the scoring and generation code paths, then reran the exact operating point behind the published claim with the fix genuinely active:

Both deltas are within noise. **No correction to the published headline numbers in Table 1 was needed**, but we disclose the bug and the check rather than silently leaving it unmentioned: the honest reading is that the bug was real, the relation gate’s isolated over-firing metric (0.97 $\rightarrow$ 0.67) does not translate into a materially stronger matched-RippleEdits preservation score at this operating point, and the compositional-preservation weakness §5.4 already names as an open limitation is not solved by activating the gate here. A separate, unrelated capacity bug in the product-key allocator (fixed on the same audit pass) is verified not to have corrupted any published number, because the single-edit-isolated protocol never approached the affected capacity ceiling; see the source addendum for the full accounting.

9 Conclusion

The result we did not expect to report is that a one-line static policy is provably optimal on all 1,689 queries we tested, because abstention can never be correct when every question in a counterfactual benchmark targets an edited fact. This is a claim about the benchmarks, not about any one system: the entire SERAC-descendant architecture family is evaluated on suites that structurally cannot re-

ward its central component, and reported routing gains on them measure how much harm a fixed gate was doing rather than any decision quality. The remedy is not a better router but a benchmark containing queries for which the correct answer is “this edit does not apply.”

INLAY, the system we built to obtain this result, is independently useful: freezing the weights and placing edits in an external addressable memory buys writes roughly $1600\times$ cheaper than gradient-based editing, exact deletion, and locality by construction, at the cost of reciting facts rather than reasoning with them. It is not universally best — WISE wins on Qwen2.5-7B CounterFact, and RAG wins on rigorously matched RippleEdits — and we report both without qualification, alongside a self-audit that found and checked two bugs in our own gating machinery rather than presenting numbers uncritically.

Reproducibility. All numbers come from logged runs. Splits are subject-disjoint with seed 0; the held-out dataset is never touched during training of any learned component. Code, per-experiment write-ups including negative results, and the figure generator are released with this paper.

Limitations

Recitation, not reasoning. Logit-space playback reproduces a stored answer; it cannot combine that answer with anything else, because by the final decoding step no reasoning remains to influence. This is structural to INLAY specifically, not to the paper’s central claim about benchmark coverage.

Scope precision on hard negatives is INLAY’s real open problem, and Table 1 is the sourced evidence for it.

Small cells. MQuAKE-CF slices are $n=63$. A +6.4-point structured-mode result rests on four discordant pairs, where the smallest attainable exact McNemar p is 0.125 — suggestive, not established.

One retrieval stack. A single encoder and one cross-encoder verifier throughout; sensitivity to those choices is untested.

A scale anomaly we diagnose but do not fix. WikiUpdate is 4.38 points *worse* at 7B than at 1.5B. It is not an editing failure: the larger model refuses to answer far more often on noisy evidence (46.88% vs 29.38%; 91.11% vs 66.67% where retrieval is wrong), and substring accuracy scores a

refusal identically to a wrong answer. The larger model is more accurate whenever it commits — it simply commits less. This is a property of the metric, and it implies scale comparisons on noisy-retrieval benchmarks systematically understate larger models.

The missing benchmark condition remains unbuilt. We argue in §7 that a benchmark with queries against a non-edited index would give routing measurable headroom for the first time, but we did not construct or run it. Our claim is limited to what current benchmarks can measure, not to what a corrected benchmark would show.

Coverage gaps in our own comparison table. No MEMIT, WISE, GRACE, or AlphaEdit results exist for Mistral-7B on CounterFact beyond the gap-fill runs reported in Appendix A, and no method besides base, RAG, and INLAY has been run through the matched-manifest RippleEdits protocol. These are open, not silently assumed favorable.

Ethics Statement

Deletion and the right to be forgotten. The central structural property of INLAY is that removing an edit is removing a row: the correction is genuinely and verifiably gone, with no residual trace in model weights. Gradient-based editing methods do not have this property — once a fact is installed by a weight update, removing it again (“unlearning”) is an open research problem, and current unlearning methods offer no comparable guarantee of completeness. This distinction is directly relevant to right-to-be-forgotten and GDPR-style deletion requirements for deployed language models: an external, addressable memory architecture is a plausible technical substrate for a system that must support real attestable deletion, in a way that direct weight editing currently is not. We surface this as a potential positive application, not as a claim that INLAY as built meets any specific regulatory standard.

Dual use: memory as an injection surface. The same property that makes deletion easy makes injection easy in the other direction. An external memory that can silently override a frozen model’s output at decode time is also a mechanism for injecting misinformation into a deployed system, if the memory store itself is not access-controlled. Unlike a weight update, which requires retraining infrastructure and typically leaves an audit trail, a

write to an addressable memory store is fast and, absent deliberate logging, easy to make invisible. Any deployment of this architecture should treat the memory store itself as a security-critical component with the same access controls one would apply to the model weights it sits beside.

Locality is conditional on gate correctness. We describe locality as holding “by construction” when the gate does not fire, and this is true bit-for-bit in that case. It is not a guarantee that the *answer* the gate returns is correct: locality-by-construction protects unrelated queries from a firing gate, but a query that should be answered from parametric knowledge and is misrouted to a stale or incorrect stored answer will confidently return that wrong answer with no signal to the user that anything is amiss. §5.4 and §8 quantify how often this misrouting occurs in our own system; a deployment should not read “locality by construction” as “correctness by construction.”

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A Full Edit-Accuracy Tables

Complete ES (edit success) / PS (paraphrase success) / NS (neighborhood specificity, i.e. locality) / harmonic-mean-score / per-edit write-cost tables for CounterFact and zsRE, across all four model families we ran. Source: `outputs/full-results-audit.md`, Parts A and B, and the audit addendum (2026-08-19), §C. Dashes mark a metric not reported for that run in the source logs.

A.1 CounterFact

A.2 zsRE

A.3 Matched-manifest RippleEdits (detail)

Method	ES	PS	NS	Score	Write (s)
<i>GPT-2-XL, n = 100 (single-edit final leaderboard)</i>					
base	0.00	–	1.00	0.00	0.0
RAG	1.00	–	1.00	1.00	0.0
finetune	1.00	–	0.25	–	34.7
ROME	0.20	–	0.00	–	15.0
MEMIT	0.40	–	0.917	–	10728.5 ^a
INLAY	1.00	–	1.00	–	0.152
<i>GPT-J-6B, N = 2000 unless noted</i>					
base (N=1000)	0.004	0.003	1.00	0.0051	–
RAG (N=1000)	0.855	0.464	0.4608	0.546	–
ROME	0.988	0.7568	0.699	0.797	8.60
MEMIT	0.458	0.2672	0.972	0.4314	26.30
WISE	1.00	0.4415	0.999	0.7032	14.76
GRACE	0.00	0.006	1.00	0.00	17.35
AlphaEdit	0.4682	0.2963	0.982	0.4595	12.12
INLAY (N=5000)	0.9976	0.8711	0.826	0.8926	load 147.6
<i>Qwen2.5-7B, N = 2000 unless noted</i>					
base (N=5000)	0.0106	0.0108	1.00	0.016	–
RAG (N=5000)	0.7983	0.4933	0.4122	0.5258	–
ROME	0.997	0.5538	0.9487	0.7767	9.30
WISE	1.00	0.8553	1.00	0.9466	27.1
GRACE	0.00	0.0055	1.00	0.00	6.8–7.0
INLAY (N=5000)	1.00	0.8776	0.8232	0.8944	load 7.87
MEMIT ^b	eff_prob 1.0 / par_prob 0.981 (dualmetric only)				34.3
AlphaEdit ^b	eff_prob 1.0 / par_prob 0.984 (dualmetric only)				16.2
<i>Mistral-7B-v0.3, N = 2000 unless noted</i>					
ROME	0.2846	0.2364	0.9211	0.3398	9.04
MEMIT ^c	0.4218	0.2509	0.9466	0.4047	–
WISE ^c	0.9962	0.3097	1.00	0.5733	10.2
INLAY (N=5000)	1.00	0.8895	0.8257	0.8995	–

Table 4: CounterFact, all model families. ^aDominated by a one-time 3-hour covariance precompute (cached); real edit cost is seconds. ^bMEMIT/AlphaEdit Qwen numbers exist only in the “dualmetric” prob-success framing, not the EasyEdit-native harmonic-mean column used for every other row; not directly comparable to INLAY’s 0.8944. ^cMistral gap-fill runs, addendum §C; no GRACE or AlphaEdit results exist for Mistral CounterFact.

Method	ES	PS	NS	Score	Write (s)
<i>GPT-2-XL, n = 100</i>					
base	0.2058	0.2063	1.00	0.2801	0.0
RAG	0.9383	0.8472	0.88	0.8869	0.0
ROME	1.00	0.795	0.9564	0.9082	3.78
MEMIT	0.2808	0.2436	0.9961	0.346	4.37
INLAY (v3)	0.96	0.96	1.00	0.973	0.34
<i>GPT-J-6B, N = 2000</i>					
base	0.2248	0.2146	1.00	0.2968	0.0
RAG	0.9069	0.8375	0.8855	0.8757	0.0
ROME	0.9951	0.9412	0.8816	0.937	10.23
WISE	0.9987	0.9892	1.00	0.9959	17.28
GRACE	0.0126	0.0013	1.00	0.0035	18.8–18.9
MEMIT	0.6146	0.5037	0.9951	0.6497	26.29
AlphaEdit	0.6345	0.5265	0.9966	0.6698	12.21
INLAY	1.00	1.00	1.00	1.00	14.20
<i>Qwen2.5-7B, N = 2000</i>					
base	0.3089	0.3045	1.00	0.3989	0.0
RAG	0.9318	0.8554	0.83	0.8703	0.0
ROME	0.9903	0.953	0.9828	0.9751	10.58
WISE	0.9991	0.9971	0.9998	0.9987	31.12
GRACE	0.0504	0.0301	1.00	0.0555	8.99
MEMIT	0.6113	0.57	0.9925	0.6822	34.80
AlphaEdit	0.6133	0.5689	0.9922	0.6824	15.32
INLAY	0.9942–0.9999	0.9949–0.9999	1.00	0.9963–0.9999	13.5–13.7
<i>Mistral-7B-v0.3, N = 2000</i>					
ROME	0.6306	0.5972	0.9856	0.7018	12.50
MEMIT ^a	0.6434	0.6178	0.9935	0.7178	19.6
WISE ^a	0.9481	0.9258	1.00	0.9570	12.7
INLAY	0.9999	0.9999	1.00	0.9999	13.69

Table 5: zsRE, all model families. ^aMistral gap-fill runs, addendum §C; no GRACE or AlphaEdit results exist for Mistral zsRE.

Model	Method	Propagation	Preservation
GPT-J-6B	base	0.0258	0.1704
GPT-J-6B	RAG	0.4566	0.2761
GPT-J-6B	ROME	0.1655	0.0633
GPT-J-6B	WISE	0.4774	0.0725
GPT-J-6B	AlphaEdit	0.1704	0.0819
GPT-J-6B	INLAY	0.3126	0.0507
Qwen2.5-7B	base	0.0425	0.3716
Qwen2.5-7B	RAG	0.4533	0.4076
Qwen2.5-7B	ROME	0.1646	0.1067
Qwen2.5-7B	WISE	0.0505	0.3254
Qwen2.5-7B	AlphaEdit	0.2060	0.0928
Qwen2.5-7B	INLAY	0.3458	0.1699

Table 6: Propagation/preservation detail behind the aggregates in Table 1. Source: the audit addendum (2026-08-19), §B.